



Today's agenda

↳ max subarray sum

↳ minimum swaps



AlgoPrep



Q) Given n elements, return max subarray sum of len: k .

$k=3$

ex: arr[9]: 2 4 3 7 9 8 5 4 3

9

14

19

24

22

17

12

= 24

Idea 1

↳ nested loop

T.C: $O(n \cdot k)$

S.C: $O(1)$

Idea 2

↳ using Prefix sum.

$k=3$

arr[9]: 2 4 3 7 9 8 5 4 3

pf[9]: 2 6 9 16 25 33 38 42 45

SP EP Sum ans :->

0 2 9 9

1 3 14 14

2 4 19 19

.

1

T.C: $O(n)$

S.C: $O(n)$



11:idea3

$k=3$
arr[9]: 2 4 3 7 9 8 5 4 3

s	e	Sum	ans:-∞
0	2	9	9
1	3	$9+7-2=14$	14
2	4	$14+9-4=19$	19
3	5	$19+8-3=24$	24



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//Pseudo code

```
int subsumk (int arr[], int k){  
    int sum=0;  
    for(int i=0; i<k; i++){ //O(k.r)  
        sum = sum + arr[i]  
    }  
  
    int ans = sum;  
    int s = 1;  
    int e = k;  
    while (e <= n) {  
        sum = sum + arr[e] - arr[s-1]  
        ans = max(ans, sum);  
        s++;  
        e++;  
    }  
  
    return ans;  
}
```

T.C: $O(n)$

S.C: $O(1)$



k=3

arr[]: 2 4 3 7 9 8 5 4 3

```
int subsumk (int arr[], int k){
```

```
    int sum=0;
```

```
    for(int i=0; i<k; i++){ //o(k-1)
```

```
        sum = sum + arr[i]
```

```
    }
```

```
    int ans = sum;
```

```
    int s = 1;
```

```
    int e = k;
```

```
    while (e <= N) {
```

```
        sum = sum + arr[e] - arr[s-1]
```

```
        ans = max(ans, sum);
```

```
        s++;
```

```
        e++;
```

```
    }
```

```
    return ans;
```

Sum = 9

Ans = 9 14 19

s

e

Sum

1

3

9+7-2=14

2

4

14+9-4=19

.

/

AlgoPrep



Q) minimum swaps

↳ Given an array of integers and an integer B , find and return the minimum number of swaps required to bring all the numbers less than or equal to B together.

ex: $arr[7]: \{1, 12, 10, 3, 14, 10, 5\}$ $B: 6$

$\{1, 3, 5, 12, 14, 10, 10\} \rightarrow 2$

ex: $arr[7]: \{1, 12, 3, 10, 14, 10, 5\}$ $B: 6$

↳ 1

$B=6$

//idea 1

$arr[7]: \{1, 12, 10, 3, 14, 10, 5\}$

2

2

2

2

2

ans = 2

B=7



arr[10]: {⁰1¹2²6³3⁴8⁵13⁶15⁷12⁸4⁹5}

window length: 5

s	e	Count $\leq B$	req swap	ans
0	4	3	$5-3=2$	2
1	5	$3+0-1=2$	$5-2=3$	2
2	6	$2+0-0=2$	$5-2=3$	2
3	7	$2+0-1=1$	$5-1=4$	2

⋮



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11Pseudo code

```
int minSwaps (int arr[N], int B){  
    int K=0;  
    for(int i=0; i<N; i++){  
        if(arr[i]<=B){K++;}  
    }  
}
```

T.C: $O(N)$

S.C: $O(1)$

```
    int c=0;  
    for (int i=0; i<K; i++){  
        if (arr[i]<=B){c++;}  
    }  
    int ans = K - c;  
    int s = 1;  
    int e = K;
```

```
    while (e<N){  
        if (arr[e]<=B){c++;}  
  
        if (arr[s-1]<=B){c--;}  
        ans = min (ans, K-c);  
        s++;  
        e++;  
    }
```

```
    return ans;
```